

Q.1. Answer the following Question

1. What is Entity Relationship Diagram?
2. What are attributes in ER Model? Explain Multivalued and Derived attribute?



1. What is Entity Relationship Diagram?



An **Entity Relationship Diagram (ERD)** is a visual representation of the relationships between entities in a database. It is a crucial part of database design and is used to outline the structure of a database, including how entities (tables) interact with one another. ERDs help database designers and stakeholders understand the system's data requirements and how data will be organized.

Key Components of an ERD:

1. Entities:

- Represented by rectangles, entities are objects or things in the real world that have a distinct existence. In a database context, entities typically correspond to tables (e.g., Customer, Order, Product).

2. Attributes:

- Attributes are the properties or characteristics of an entity, represented by ovals connected to their respective entities. For example, a Customer entity might have attributes such as CustomerID, Name, and Email.

3. Relationships:

- Represented by diamonds or lines connecting entities, relationships describe how entities interact with each other. Relationships can be classified as:

- **One-to-One (1:1):** One record in Entity A is related to one record in Entity B.

- **One-to-Many (1**

): One record in Entity A can relate to many records in Entity B, but a record in Entity B relates to only one record in Entity A.

- **Many-to-Many (M**

): Many records in Entity A can relate to many records in Entity B.

4. Cardinality:

- Cardinality defines the numerical relationships between entities, indicating how many instances of one entity can or must be associated with instances of another entity. It is usually expressed using symbols like 0, 1, or N (for many).

5. Keys:

- Primary keys and foreign keys can be indicated in an ERD to show the unique identifiers for entities and the relationships between them.

Purpose of an ERD:

- **Database Design:** ERDs are used during the database design process to help visualize the data model before implementation.
- **Communication Tool:** They serve as a communication tool among stakeholders, including database designers, developers, and non-technical team members, to ensure a clear understanding of data relationships.
- **Documentation:** ERDs provide documentation that can be useful for future reference, maintenance, and updates to the database system.

Example of an ERD:

Imagine a simple ERD for a library system that includes three entities: **Book**, **Author**, and **Publisher**.

- **Entities:**

- **Book:** Attributes could include BookID, Title, PublishedYear.
- **Author:** Attributes could include AuthorID, Name, Nationality.
- **Publisher:** Attributes could include PublisherID, Name, Location.

- **Relationships:**

- A **Book** can have one or more **Authors** (Many-to-Many relationship).
- A **Book** is published by one **Publisher** (One-to-Many relationship).

2.What are attributes in ER Model? Explain Multivalued and Derived attribute?



In the Entity-Relationship (ER) model, **attributes** are the properties or characteristics that define an entity. Each attribute holds specific data related to the entity and helps describe its state. For example, if you have an entity called **Customer**, it might have attributes such as **CustomerID**, **Name**, **Email**, and **Phone Number**.

Types of Attributes in the ER Model:

1. Simple Attribute:

- A single-valued attribute that holds a single value for each entity instance.
- Example: FirstName in a **Customer** entity.

2. Composite Attribute:

- An attribute that can be divided into smaller sub-attributes, which represent more basic attributes with independent meanings.
- Example: Address can be broken down into Street, City, State, and Zip Code.

3. Derived Attribute:

- An attribute whose value can be derived or calculated from other attributes in the database. It does not need to be stored because it can be computed from other attributes.
- Example: An attribute Age can be derived from a DateOfBirth attribute by calculating the difference between the current date and the date of birth.

4. Multivalued Attribute:

- An attribute that can hold multiple values for a single entity instance. This means that a single entity can have a set of values for this attribute.
- Example: In a **Student** entity, the attribute PhoneNumbers might allow multiple phone numbers for a single student.

Multivalued Attribute

- **Definition:** A multivalued attribute is an attribute that can have multiple values for a single entity. It is represented in an ER diagram by a double oval.
- **Example:**

- For a **Student** entity, the EmailAddresses attribute could be multivalued if a student can have multiple email addresses.
- In this case, you would represent EmailAddresses as a multivalued attribute in the ERD.

Derived Attribute

- **Definition:** A derived attribute is an attribute that is calculated or derived from other attributes within the same entity or from related entities. It is represented in an ER diagram by a dashed oval.
- **Example:**
 - For an **Employee** entity, the YearsOfService attribute could be derived from the DateOfJoining attribute. Instead of storing YearsOfService, it could be computed based on the current date and the date the employee joined the company.